Unit 8. Point Estimation: MLE and OLS

Bias, consistency, efficiency; Maximum Likelihood; OLS regression

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lec. 17–18; OLS framing follows Stock & Watson, *Introduction to Econometrics*.

Roadmap

1. Estimators and their sampling distribution
2. Bias, consistency, efficiency
3. Maximum Likelihood Estimation
4. The simple linear regression model
5. The OLS estimator
6. Sampling properties of OLS
7. OLS as MoM and as MLE
8. Wrap-up

The estimation problem

Setup. Random sample $X_1, \ldots, X_n$ from $\{f(\cdot \mid \theta) : \theta \in \Theta\}$. The true θ_0 is unknown.

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Mean Squared Error

$$\mathrm{MSE}(\widehat{\theta}) = \mathrm{Bias}^2 + \mathrm{Variance}.$$

Three properties of an estimator

Definitions

- Unbiased: $\mathbb{E}_{\theta_0}[\widehat{\theta}] = \theta_0$.
- Consistent: $\widehat{\theta} \xrightarrow{p} \theta_0$.
- Efficient: smallest variance among unbiased competitors.

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Independence of the properties.

- Unbiased but not consistent: $\widehat{\theta} = X_n$ (last observation).
- Biased but consistent: $\widehat{\theta} = \max_i X_i$ for $X_i \sim U[0, \theta_0]$.

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Sample mean. $\mathrm{SE}(\bar{X}_n) = \sigma/\sqrt{n}$; estimated as $S_n/\sqrt{n}$. **Sample variance:** n **vs** $n-1$**.**

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \bar{X}_n)^2 \text{ biased,} \qquad S_n^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X}_n)^2 \text{ unbiased.}$$

The -1 adjusts for the degree of freedom used to estimate $\bar{X}_n$.

Maximum Likelihood: principle

Definitions

$$\mathcal{L}_n(\theta) = \prod_{i=1}^{n} f(X_i \mid \theta), \qquad \ell_n(\theta) = \sum_{i=1}^{n} \log f(X_i \mid \theta).$$

$$\widehat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta \in \Theta} \ell_n(\theta).$$

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Why it works.

- Population log-likelihood $\mathbb{E}_{\theta_0}[\log f(X \mid \theta)]$ is maximised at $\theta = \theta_0$ (Jensen).
- LLN: $\ell_n(\theta)/n \xrightarrow{p} \mathbb{E}_{\theta_0}[\log f(X \mid \theta)]$.
- Hence $\widehat{\theta}_{\mathrm{MLE}} \xrightarrow{p} \theta_0$ under regularity.

Example: MLE depends on the observed sample

Suppose there are two possible samples, $x^{(1)}$ and $x^{(2)}$, and two possible parameter values, θ_1 and θ_2.

$L_n(\theta)$	$x^{(1)}$	$x^{(2)}$
θ_1	$\frac{1}{4}$	$\frac{1}{8}$
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Key point: the MLE is computed from the observed sample. If the sample changes, the likelihood function changes, and the resulting estimate may also change.

Examples: normal and uniform

Normal $\mathcal{N}(\mu, \sigma^2)$.

$$\widehat{\mu}_{\mathrm{MLE}} = \bar{X}_n, \qquad \widehat{\sigma^2}_{\mathrm{MLE}} = \frac{1}{n}\sum_i (X_i - \bar{X}_n)^2.$$

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$\mathbb{E}[\widehat{\theta}_{\mathrm{MLE}}] = \frac{n}{n+1}\theta_0$ – biased, but lower MSE than MoM at moderate n.

Properties of the MLE

Under regularity (smooth density, identifiable θ, finite Fisher information):

- Consistent: $\widehat{\theta}_{\mathrm{MLE}} \xrightarrow{p} \theta_0$.
- Asymptotically normal: $\sqrt{n}(\widehat{\theta}_{\mathrm{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0,\, I(\theta_0)^{-1}\right)$.
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Caveats.

- May be biased in finite samples (e.g. uniform).
- Often hard to compute (numerical optimisation).
- Sensitive to misspecification of f.

Simple linear regression (Stock & Watson)

We observe a random sample $\{(X_i, Y_i)\}_{i=1}^{n}$. The model is

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Three Least-Squares Assumptions:

(LSA1) $\mathbb{E}[u_i \mid X_i] = 0$ zero conditional mean / *exogeneity*.

(LSA2) (X_i, Y_i) are i.i.d.

(LSA3) $\mathbb{E}[X^4], \mathbb{E}[Y^4] < \infty$ large outliers unlikely.

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Optional extras:

(LSA4) Homoskedasticity: $\mathrm{Var}(u_i \mid X_i) = \sigma_u^2$.

(LSA5) Conditional normality: $u_i \mid X_i \sim \mathcal{N}(0, \sigma_u^2)$.

What does (LSA1) buy us?

$\mathbb{E}[u \mid X] = 0 \;\Rightarrow$

$$\mathbb{E}[u] = 0, \qquad \mathrm{Cov}(X, u) = 0, \qquad \mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x.$$

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Slogan

Under (LSA1), the regression line is the population CEF, and the slope β_1 measures the causal-style ceteris-paribus effect of X on $\mathbb{E}[Y \mid X]$.

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Violating (LSA1) is the central concern of econometrics. Examples: omitted variables, measurement error, simultaneity (recall Unit 4).

Deriving OLS

Objective. Minimise the residual sum of squares:

$$(\widehat{\beta}_0, \widehat{\beta}_1) = \arg\min_{b_0, b_1} \sum_{i=1}^{n} (Y_i - b_0 - b_1 X_i)^2.$$

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Normal equations.

$$\sum_i (Y_i - \widehat{\beta}_0 - \widehat{\beta}_1 X_i) = 0, \quad \sum_i X_i (Y_i - \widehat{\beta}_0 - \widehat{\beta}_1 X_i) = 0.$$

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Solution.

$$\boxed{\widehat{\beta}_1 = \frac{s_{XY}}{s_X^2} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}, \qquad \widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X}.}$$

The sample analogue of $\mathrm{Cov}(X, Y)/\mathrm{Var}(X)$ from Unit 6.

Mechanical (algebraic) properties

Define $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$ and $\widehat{u}_i = Y_i - \widehat{Y}_i$.

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1. $\sum_i \widehat{u}_i = 0$ (residuals sum to zero).
2. $\sum_i X_i \widehat{u}_i = 0$ (orthogonality to the regressor).
3. $\bar{Y} = \widehat{\beta}_0 + \widehat{\beta}_1 \bar{X}$ (line passes through $(\bar{X}, \bar{Y})$).
4. Sums of squares: $\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}$; $R^2 = \mathrm{ESS}/\mathrm{TSS}$.

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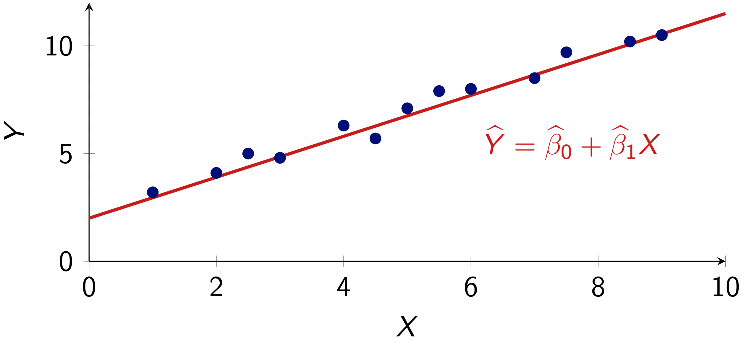
These are pure algebra: they hold for any data set, with no need for any LSA.

OLS line: a picture



The OLS line minimises the sum of squared vertical distances of the data points to the line.

Sampling properties under (LSA1)–(LSA3)

Theorem

Under (LSA1)–(LSA3) and $\mathrm{Var}(X) > 0$:

1. Unbiasedness: $\mathbb{E}[\widehat{\beta}_1] = \beta_1$.
2. Consistency: $\widehat{\beta}_1 \xrightarrow{p} \beta_1$.
3. Asymptotic normality:

$$\sqrt{n}\big(\widehat{\beta}_1 - \beta_1\big) \xrightarrow{d} \mathcal{N}\Big(0, \ \frac{\mathrm{Var}\big[(X - \mu_X)u\big]}{[\mathrm{Var}(X)]^2}\Big).$$

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Inference toolkit: replace the asymptotic variance by its sample analogue $\Rightarrow$ heteroskedasticity-robust standard errors.

This is what `regress, robust` (Stata) and `coeftest(..., vcov=vcovHC)` (R) compute by default.

Gauss–Markov: BLUE under (LSA4)

Theorem

Add (LSA4) homoskedasticity: $\mathrm{Var}(u \mid X) = \sigma_u^2$.
Then OLS is the Best Linear Unbiased Estimator: smallest variance among all linear unbiased estimators.

Conditional variance simplifies:

$$\mathrm{Var}(\widehat{\beta}_1 \mid X) = \frac{\sigma_u^2}{\sum_i (X_i - \bar{X})^2}.$$

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Practical recommendation. Use robust standard errors by default. Homoskedasticity-only SEs are smaller *when* (LSA4) holds but invalid when it doesn't. Robust is always valid under (LSA1–3).

Three principles, one estimator

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OLS as MLE. Add (LSA5) conditional normality. Then $Y \mid X \sim \mathcal{N}(\beta_0 + \beta_1 X, \sigma_u^2)$ and

$$\ell_n = -\tfrac{n}{2}\log(2\pi\sigma_u^2) - \tfrac{1}{2\sigma_u^2}\sum_i (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Maximising ℓ_n in $(\beta_0, \beta_1) \;\Leftrightarrow\;$ minimising RSS = OLS.

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Maximising ℓ_n in $(\beta_0, \beta_1) \Leftrightarrow$ minimising RSS = OLS.

Punchline

All three estimation principles — least squares, MoM, MLE — yield the same OLS coefficient. Each adds a different justification.

Putting it together: a worked example

A student tracks her commute time Y_i (minutes) and number of buses $X_i \in \{0, 1, 2\}$ over $n = 60$ days. Sufficient statistics:

$$\bar{X} = 0.95, \quad \bar{Y} = 28, \quad s_X^2 = 0.45, \quad s_{XY} = 2.7.$$

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Reading. Under (LSA1)–(LSA3), $\mathbb{E}[\text{commute} \mid \text{buses} = x] = 22.3 + 6x$.

Inference (CI, t-test) is the subject of Unit 10.

Take-aways

1. Estimators have a sampling distribution; care about bias, consistency, and efficiency.
2. Method of Moments: equate sample to population moments; simple, consistent, often inefficient.
3. MLE: maximise the likelihood; consistent and asymptotically efficient under regularity.
4. Linear regression model (Stock–Watson): $Y = \beta_0 + \beta_1 X + u$ with (LSA1)–(LSA3).
5. OLS estimator: $\widehat{\beta}_1 = s_{XY}/s_X^2$, unbiased and consistent under (LSA1–3); BLUE under (LSA4); MLE under (LSA5).

Next unit

Confidence intervals and hypothesis testing — including inference on $\widehat{\beta}_1$.

Thank you

Questions?

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